

The Verification Gap: Auditing Proof Progress and Repair on IMO 2026

Full list of corrections since the reviewed version

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This list accompanies the camera-ready version of the paper (MATH-AI 2026, the 6th Workshop on Mathematical Reasoning and AI at NeurIPS 2026). Appendix Q of the paper states which version the referees read, summarizes the corrections that change what a reader should conclude, and reproduces the reviewed abstract and TLDR. Below is one row per corrected sentence; in the paper’s source each corrected sentence carries the same identifier. Section numbers in the “Reviewed” part of a row are those of the reviewed version; appendix letters are the camera-ready’s.

Rows. One row per corrected-sentence identifier: the reviewed wording (section numbers are the reviewed version’s), what replaces it and where, and who raised it; the evidence for each row is the stored records named in the section it points to. Sources: a referee by OpenReview identifier (q51H, qUaw, YmcM, s3f5); “audit” for the internal model-run challenge audit of 11 September 2026; “panel” for the model-run camera-ready panel and provenance trace of 29 September 2026 (Appendix P). Identifiers Q38 to Q40 have no row: Q38 is the OpenReview metadata (Appendix Q of the paper), and Q39 and Q40 concern no text of the paper.

- Q1 Reviewed (abstract, §8, App. L, TLDR): the response “carries a field that does not” drop the objection; “The response carried the objection; the number did not.” Now: the completeness flag is nested between the cuts at 6 and 7 and departs from a 7 only on GPT-5.6-Sol responses graded 6 (4 of 64 in the first run, 2 in the re-query); the prompt names “complete” without defining it; the tagline is deleted (body; App. L.7). Source: q51H.
- Q2 Reviewed (abstract, §8, App. L): “no pass mark on the 0–7 scale reaches the operating point of the completeness flag”; “no mark recovers the flag’s operating point”. Now: a trade stated per run: in the first run the strict mark gives (1, 9) broken accepted and sound rejected and the flag (1, 6); in the re-query (1, 8) against (2, 7), so neither dominates (body; App. L.7). Source: s3f5, q51H; audit.
- Q3 Reviewed (abstract, §8): GPT-5.6-Sol told fatal and harmless edits “apart at chance (0.50 against 0.94 for the other)”. Now: its scores had no variance across the fatal and harmless edits in the first run (all 6; one harmless edit at 7 in the re-query), it graded the harmless edit below the original on 7 of 8 bases, and the area under the curve, ties counted one half, is reported as a repair-convention quantity (body; App. L.7). Source: q51H.
- Q4 Reviewed (abstract): “over 24 published single-hunk mutants, 16 with an executable check, . . . rejects 9 of 16 sound ones”. Now: 32 texts; broken means fatal or gap (16), sound means an original or a harmless edit (16), sound only under the repair-tolerant convention; checks exist on the fatal and harmless edits; “single-edit” replaces “single-hunk” (body; App. L.3). Source: audit.
- Q5 Reviewed: the harmless class also called “made ugly” and “cosmetic”; “every proof we made false”. Now: one name, harmless, a false line nothing later uses; a fatal edit leaves the argument invalid as written, not necessarily the theorem false; the adjudication status “false display” and that adjudication’s broader repair-tolerant convention are distinguished from it (body; App. L.3, K). Source: panel.
- Q6 Reviewed (§8, Table 1 caption, App. L): post-hoc reads unlabelled; “the pre-registered gate list”, “the pre-registered counts”. Now: every read is labelled prespecified or post hoc, and “prespecified (self-timestamped)” replaces “pre-registered” (body; App. L.1). Source: panel.
- Q7 Reviewed: “the protocol timestamped beforehand” (§8); “Under a protocol fixed before any call” (§6); “timestamped before every mutant call” (App. L); “written and timestamped . . . before the first call” (App. J); “timestamped before the first call”

- (App. K). Now: the planted protocol was committed at launch, 20 s after the earliest mutant request, and never edited, every mutant request carrying its recorded digest, after 3 smoke-test screenings; the witness-discovery and reference-adjudication protocols are prespecified and self-timestamped, and their first commits follow their first calls (body; App. L.1, J, K; the body now states the witness-discovery protocol’s status only as prespecified and points to App. J). Source: qUaw, who cited the timestamp; panel.
- Q8 Reviewed: “we planted . . . ours to name” (§8); “The plantings are author-made” and “assigned by us” (App. L); App. P named only scripts and prose. Now: the study design is the author’s; AI agents (a coordinating Claude Fable 5.1 agent and three Claude Fable 5.1 subagents) implemented it, writing the specification, edits, dependency arguments, intended labels and checks, and the author read no individual edit before the gates ran (abstract; body; App. L.1, P). Source: panel (traced from the session records); design attribution confirmed by the author.
- Q9 Reviewed (§8, App. L): “the 16 fatal and harmless edits carry an executable check”, with no statement of what a check establishes. Now: a check certifies that the planted statement is false (for P5, that the expansion is not an identity), not that a consumer fails; the deletions have none; a dated P5 release note (body; App. L.3). Source: panel.
- Q10 Reviewed: “Setting the four disputed cases aside . . . a fifth edit open to the same repair” and “Four named cases are disputed” (§8), “Four severity labels are disputed” (App. L), beside “five are contested” (§9). Now: five disputed labels, named with reasons and each rule’s decision; the exclusion ladder for the pass mark, the strict mark, the flag (per run) and the witness gate; the convention reversal without them; and the repair criterion’s reach to the undisputed fatal edits (body; App. L.4, L.7). Source: YmcM; audit.
- Q11 Reviewed (§8): the only worked fatal example was the P5 sign flip, “writing $(t - y)^2 + 4(t + c)(b - c)$ where the expansion gives $(t - y)^2 + 4(t + c)(c - b)$ ”. Now: an undisputed example generated from the stored files (p6-gpt56-ours-F) is shown in Figure 1, and the P5 sign flip is listed as disputed, with the reason (App. L.4). Source: YmcM; panel.
- Q12 Reviewed (App. L): “Eight base texts that every gate accepts”. Now: accepted at selection, by model verdicts; four decision variants reject the Kimi K3 P1 original, which carries an unused false remark, and relabelling it leaves the convention reversal intact (App. L.2). Source: audit.
- Q13 Reviewed: “a human read the six GPT-5.6-Sol proofs against” Chen’s notes (§2); “human-read against the published solutions” (App. L). Now: the author compared the six proofs with the notes and found them consistent (not a line-by-line or expert check; no written record); a Claude model’s step-by-step comparison of 5 September 2026 is disclosed; a model double-check of 29 September 2026 found the six GPT-5.6-Sol bases consistent with the notes; acceptance of the bases is model verdicts, from the base’s own lab for five (body; App. L.2, P). Source: q51H, qUaw, YmcM (no human expert anywhere); panel.
- Q14 Reviewed: “closure and witness are separately prompted” (Table 1 caption); “proposers and graders saw only the rubric and the text” and “what is matched is the number of first-stage calls and the checking rule, not tokens or time” (App. J). Now: the witness and closure prompts carry the problem’s marking scheme, which the grade prompt forbids; comparisons are labelled reference-guided against reference-free; the arms are matched in neither information nor instruction; the grading prompt, the witness instructions and the P1 and P5 rubrics are printed (body; App. B, J, L.6). Source: qUaw; audit.
- Q15 Reviewed: not stated. Now: both P1 collisions are named and scoped (the fatal-edit collision bears only on the comparison with the pass mark; the gap-edit collision is the witness gate’s whole broken-text margin over the strict mark and the flag); four edits that re-plant natural defects, and who saw which rubric or ledger, are stated; a later witness run without the scheme is in App. S (body; App. L.3, L.6). Source: audit.
- Q16 Reviewed (abstract): “Referring only what two graders dispute costs 4/32 under an obligation-level witness rule, which passes 0/16 broken proofs but refuses 1/16 sound ones, against 10/32 and 1/16 for the flag”. Now: removed from the abstract; routing is given for every rule on both axes, with populations and runs (body; App. L.7). Source: s3f5.
- Q17 Reviewed: “Escalation is where a remedy pays”, “costs”, “buys” (§8), “spend the human budget on disagreement” (§9), “One escalation is a stipulated unit of effort” (App. L). Now: the witness rule is a comparator; a referral is a count of texts sent to a person, not timed; and the operator advice routes disagreements under a strict rule and treats agreement as no certificate (body; App. L.7). Source: YmcM, qUaw.
- Q18 Reviewed (App. L): the witness gate is “positive” (“fewer false accepts than the grade at pass mark 5, no more false rejects”). Now: both denominators, 0 of 8 originals as prespecified and 5 of 16 sound texts against the pass mark’s 0 (App. L.7). Source: audit.
- Q19 Reviewed: the witness rule’s one sound refusal was not explained. Now: it is the planted copy of the Nemotron P5 display, a correct refusal as written (body; App. L.7). Source: panel.
- Q20 Reviewed (abstract, §8): “quoting the edited passage in 7/13” as the headline. Now: at least one grader quoted the edited passage on all 13 accepted broken texts in both runs; GPT-5.6-Sol named the planted error on every fatal edit; its own quote count is 7 by exact and 9 by tolerant anchoring (body; App. L.7). Source: panel.
- Q21 Reviewed: a gap edit as “a load-bearing proof deleted, its statement kept” (§8); the refused gaps as texts “whose deleted proof left a visible hole” (App. L); “planted edits are easier than natural ones” (§8). Now: every gap replacement contains “(standard; details omitted)”, so every deletion announces itself; the refused gaps are those at least one grader scored below 5, namely the P6 deletions and the deletion in the Claude-written P3 base, with no P1 or P5 deletion refused (a post-hoc pattern

- confounded with problem), with no mechanism claimed; “presumably easier”; unannounced deletions are untested (body; App. L.3, L.7). Source: s3f5; panel.
- Q22 Reviewed (abstract, §8): “reverses . . . on all 3 strict rules”, “strict rules” meaning two rules in one place and three in another, with no run qualifier. Now: 3 of 3 in the first run and 2 of 2 re-queried, with the rules named once, the flag’s per-lab counts given and the witness gate marked as run once; the abstract and §1 state the reversal at 7, in both runs, and §4 gives the completeness field’s, because in later runs the strict mark’s reversal always recurs and the flag’s not always (App. S); the identification claim is scoped to these two graders (abstract; body; App. L.7, S). Source: audit; panel.
- Q23 Reviewed (§6): “both recovered every documented defect (7 of 7)”. Now: 6 distinct defects on 7 texts (body; App. J). Source: audit.
- Q24 Reviewed (App. E): the labs’ verdicts “differ on 10 of 32 planted texts against 2 of 32”. Now: 10 of the 24 mutants and 0 of the 8 originals, against 2 of 24 for the pass mark (App. E). Source: audit.
- Q25 Reviewed (App. E): “Same-lab endorsement of objections is 95/96.” Now: same-lab 95/96 and other-lab 95/96, and endorsement is not a verdict that an objection identifies a defect (App. E). Source: q51H.
- Q26 Reviewed (§9): “the checkers are the objection-producing labs”, unscoped. Now: scoped to the reference adjudication and the planted study’s reference gate (body; App. K). Source: q51H.
- Q27 Reviewed: the alarm’s “4 of 4” without “retrospective” (§3); the limitations inside App. E, titled “Network-Access Audit”. Now: “retrospective except on the planted texts”; the limitations are in the body, and App. E keeps its title with a pointer and a further-limitations paragraph (body; App. E, I). Source: panel.
- Q28 Reviewed (§3): κ without scope; “Evaluator variation as apparent progress”. Now: κ 0.78–0.84 on the trajectory corpus; on the planted texts Claude Opus 5 and GLM-5.3 agree on every text (κ 1.00) yet each gives $S=2$ to most fatal edits, while GPT-5.6-Sol departs from both (κ 0.42); agreement is reliability, not validity, and S was never validated against an expert; evaluator variation can masquerade as progress and under replay rarely does (body; App. F, I). Source: qUaw.
- Q29 Reviewed: App. I, which referee q51H called “longer than most workshop submissions”, ended in survival figures equal to one for a reference with no regression. Now: a reading guide says which figures are informative; App. I keeps repeat variability, the replay null and the alarm, and the full tables stay in the supplement (App. I); it is cut by about a third, to about two pages rather than the one planned, because the replay null and the alarm’s tables stay in the PDF. Source: q51H.
- Q30 Reviewed (§1): graders “rewarded by confident proofs of false statements”, citing BrokenMath, Proof or Bluff and MathArena. Now: score inflation by model judges is attributed to Petrov et al. (2025a), convincing proofs of false statements to provers (Petrov et al., 2025b), and MathArena leaves that clause (body). Source: panel.
- Q31 Reviewed (App. P): “every number in the body and the appendices is generated . . . rather than typed by hand”; “every one of the fifteen printed references was checked”. Now: every count in the body is a generated macro under a literal check, the appendix figures are generated apart from a few counts written in words, and the added references were checked against their records (App. P). Source: panel.
- Q32 Reviewed: the prespecified per-author report was printed only for both graders together. Now: per grader, rule and base lab, for both runs (body; App. L.7). Source: qUaw, q51H.
- Q33 Reviewed: no statement that the texts are non-independent, the bases hand-picked, or the problem coverage partial. Now: 8 bases on four of the 6 problems, chosen among texts model evaluators had accepted and not a sample, whose 32 texts are not independent (body; App. L.2). Source: YmcM, s3f5, q51H, qUaw.
- Q34 Reviewed (App. L): repair language “appears in 14/32 grading responses”, unscoped. Now: 14 of 32 responses in the main gap and error descriptions (21 counting the free notes), a keyword match and not a severity coding (App. L.7). Source: panel.
- Q35 Reviewed: no licence notice; “We release the corpus, ratings, witnesses, certificates and ledger.” Now: the archive carries NOTICE.md (third-party texts redistributed for reproduction only, no licence granted; the author’s material under CC BY 4.0), and the release sentence names the repository (body; App. P). Source: panel (compliance).
- Q36 Reviewed (App. L): “added after the reviews”, “added after the v12 reviews”, “A reviewer identified a second P1 gap”, “disputed after review”, which read as the workshop’s referees. Now: these are internal model-run reviews before submission, disclosed in App. P (App. L.1). Source: panel (compliance).
- Q37 Reviewed: the build’s claim guard lacked sound-side and run checks, and one generated table was emitted but never used. Now: the claim guard is extended (tooling only, listed in the archive’s CHANGES.md); the unused generated table is still emitted and still unused; no sentence of the paper depends on either (App. P). Source: audit.
- Q41 Reviewed (§1): “several frontier models solved all six problems under thin tool loops”; “In these campaigns solving was not the bottleneck; knowing that a proof is correct was.” Now: they “were credited with all six problems by the campaign’s Claude-based verifier agents”, and the bottleneck was “not producing credited proofs but knowing that a proof is correct: the verification gap” (body). Source: panel.
- Q42 Reviewed (§1): “Closest work: reference-guided grading of final proofs validated against human grades . . . ; we differ in . . . annotating obligations rather than whole proofs”. Now: a positioning paragraph that places RefGrader (the “Mahdavi et al., 2025” of referee s3f5’s review), binary judging, ProofBench’s reference contexts, planted perturbations and inserted bugs, and a taxonomy of proof fallacies, maps the grade arm to reference-free and the witness and closure arms to reference-guided,

gives the repair-tolerant witness-plus-reference gate (App. L.7) as the nearest in-paper analogue, with its counts, noting that its convention partly builds in its zero sound rejections, and states that no head-to-head comparison was run (body). Source: s3f5; panel.

- Q43 Reviewed (§1): the contributions listed the failure catalogue, the trajectory corpus, the probes and the witness study before the planted-defect comparison. Now: in reading order, the planted study first, then convention dependence, the reference-guided comparator and the campaign audit behind the design (body). Source: q51H, qUaw, YmcM.
- QA1 Reviewed: the ledger “written by the author” (App. K), `recovery.json` “judged by the author” (App. J), the pilot “author read” (App. M), “we compared the text step by step” (App. F). Now: produced by a model under the author’s direction and reviewed by the author (App. F, J, K, M, P). Source: panel (provenance trace).
- QA2 Reviewed: not stated. Now: two ledger issues found by a model check on 29 September 2026, neither changing a recorded judgment, are disclosed (App. K). Source: panel.
- QA3 Reviewed: not stated that the pass-mark result depends on the rubric. Now: “Scope: one rubric; a rubric that caps an unresolved load-bearing error below 5 was not run in this protocol”; the instructed arm of App. S, whose no-repair instruction keeps the 5–6 band, is labelled a partial instruction control, not that rubric (body; App. S). Source: YmcM.

References

- Ivo Petrov, Jasper Dekoninck, Lyuben Baltadzhiev, Maria Drencheva, Kristian Minchev, Mislav Balunović, Nikola Jovanović, and Martin Vechev. Proof or bluff? Evaluating LLMs on 2025 USA math olympiad. *arXiv preprint arXiv:2503.21934*, 2025a.
- Ivo Petrov, Jasper Dekoninck, and Martin Vechev. BrokenMath: A benchmark for sycophancy in theorem proving with LLMs. *arXiv preprint arXiv:2510.04721*, 2025b.